



PSO201/PHY204: Assignment 4
Jan-Apr 2025

1. The expression of two spherical harmonics are given below.

$$Y_{00} = \frac{1}{\sqrt{4\pi}}; Y_{21} = -\sqrt{\frac{15}{8\pi}} e^{i\varphi} \sin \theta \cos \theta$$

Verify that the two spherical harmonics are normalized and orthogonal to each other.

2. Let $\varphi_{nlm}(r)$ denote the properly-normalized energy eigenfunctions of the coulomb potential with principal quantum number n and angular momentum quantum numbers l and m . Consider an electron in the state $\psi(r) = C[\varphi_{100}(r) + 4i\varphi_{210}(r) - 2\sqrt{2}\varphi_{21-1}(r)]$
- Find the normalization constant C .
 - What is the expectation value of E ?
 - What is the expectation value of L^2 ?
 - What is the expectation value of L_z ?
 - Write down $\psi(r, t)$ at some later time t .
3. Consider an electron trapped in a coulomb potential in the state corresponding to maximum orbital angular momentum, $l = n - 1$. Find $\langle r \rangle$ and $\langle r^2 \rangle$. Use the results to show that, for large values of n , the electron is well localized near the surface of a sphere of radius $n^2 r_0$ and has energy which is the same as that of a classical electron in a circular orbit of the same radius.
4. Show that the solution of the radial Schrodinger equation for a hydrogenic potential $V(r) = -\frac{ze^2}{r}$ in the limit $r \rightarrow \infty$ is $u(r) = e^{-\frac{Zr}{na_0}}$.
5. Find $\langle x \rangle$ and $\langle x^2 \rangle$ for the ground state of the hydrogen atom, where $r^2 = x^2 + y^2 + z^2$. What is the most probable value of r ?
6. Generalize the conversion of position space wave function to momentum space wavefunction in 3D. Find the momentum space wavefunction for the ground state of hydrogen.
7. An electron is confined in a three-dimensional infinite potential well. The sides parallel to the x-, y-, and z-axes are of length L each.
- Write the appropriate Schrodinger equation.
 - Find the ground state energy and the ground state wave function.
 - Give an expression for the number of states, having energy less than some given E .
8. A particle of mass m is constrained to move between two concentric impenetrable spherical surfaces of radii a and b . The potential in the region $a < r < b$ is zero. Find the ground state energy and normalized wave function.
9. Consider a (spinless) particle represented by the wave function $\psi = \alpha(x + y + 2z)e^{-\beta r}$ where α, β are real constants.
- What is the total angular momentum of the particle?
 - What is the expectation value of the z-component of the angular momentum L_z ?
 - If the z-component of angular momentum were measured, what is the probability that the result would be $L_z = +\hbar$?
10. A particle of mass m in three dimensions is in the potential $V(r) = 0$ for $r < a$ and $V(r) = \infty$ for $r > a$. Find the ground state energy.